

Weekly Problem 1

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Exercise 1

Part (a)

1	$p \vee q$	
2	$\neg p$	
3	p	
4	$\perp$	$\neg E, 2, 3$
5	q	$\perp E, 4$
6	q	
7	q	$R, 6$
8	q	$\vee E, 1, 3-5, 6-7$
9	$\neg p \Rightarrow q$	$\Rightarrow I, 2-8$

We then see that $p \vee q \vdash \neg p \Rightarrow q$ is a valid argument in classical logic.

Part (b)

Let $p = 1$, $q = 0$, and $r = 1$. We see that $q \vee r = 1$, so $p \Rightarrow (q \vee r) = 1$. However, $\neg p = 0$. Consequently, the argument $p \Rightarrow (q \vee r), \neg q \vdash \neg p$ is an invalid argument in classical logic.